

Sri Lanka Institute of Information Technology



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Web Based Transportation System

SE2030 - Software Engineering

B.Sc. (Hons) in Information Technology

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Test Case Designing – Web Based Transportation System

1.User Registration.

Test Case ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
TC_01	Register with valid inputs	Ensure user can log in with correct username and password.	1. User must be registered. 2. User must be on login page.	1. Open the login page. 2. Enter valid name. 3. Enter valid password. 4. Click Register.	Name: John, Email: john@mail.com, Pw: Test@123	User is successfully redirected to the home page.	“Registration successful. Please login.”	Pass
TC_02	Register with missing fields	Ensure registration fails if required fields are empty.	User is on registration page.	1. Leave one or more mandatory fields blank. 2. Click Register.	Email: (blank), Pw: Test@123	Error message “All fields are required”.	“Please fill out all required fields.”	Pass
TC_03	Register with invalid email format	Verify system rejects invalid email address.	User is on registration page.	1. Enter valid name and password. 2. Enter invalid email format (e.g., “abc@com”) 3. Click Register.	Email: user@abc, Pw: Test@123	Error message “Invalid email format”.	“Enter a valid email address.”	Pass
TC_04	Register with weak password	Ensure weak passwords are rejected.	User is on registration page.	1. Enter valid name and email. 2. Enter weak password (e.g., “123”). 3. Click Register.	Email: tom@mail.com, Pw: 12345	Error message “Password must meet complexity rules”.	(To be filled after execution)	Pass
TC_05	Register with existing email	Verify system prevents duplicate registrations.	Email already exists in system.	1. Enter valid name, existing email, and valid password. 2. Click Register.	Email: john@mail.com, Pw: Test@123	Error message “Email already registered”.	(To be filled after execution)	Pass

TC_06	Register with mismatched passwords	Ensure system validates password and confirm password.	User is on registration page.	1. Enter valid details. 2. Enter password and confirm password that do not match. 3. Click Register.	Email: anna@mail.com , Pw: Test@123, Confirm: Test@456	Error message "Passwords do not match".	"Confirm password does not match."	Pass
TC_07	Register with special characters in username	Verify username rules are enforced.	User is on registration page.	1. Enter username with invalid special characters (e.g., "@John!!"). 2. Click Register.		Error message "Invalid username format".	"Username cannot contain special characters."	Pass
TC_08	Register with valid long input	Ensure long but valid inputs are accepted.	User is on registration page.	1. Enter long valid username and strong password. 2. Enter valid email. 3. Click Register.	Long username, Long password	Registration successful, redirected to home page.	"Registration successful . Welcome!"	Pass
TC_09	Register with SQL injection attempt	Verify system is secure against SQL injection in input fields.	User is on registration page.	1. Enter malicious input (e.g., "test'; DROP TABLE users;--"). 2. Click Register.	SQL injection string	Error message shown, no system crash, input sanitized.	"Special characters not allowed."	Pass
TC_10	Register and auto-login after success	Ensure system logs in user automatically after registration.	User is not logged in.	1. Enter valid registration details. 2. Click Register.	Valid username, email, password	User registered successfully and redirected to dashboard.	"Registration successful . Logged in"	Pass

2.Ride Booking.

Test Case ID	Test Title	Description	Precondition	Test steps	Test Data	Expected Outcome	Actual Output	Status
TC - 001	Book Ride Successfully	Test successful ride booking with valid locations.	User is logged in; App is responsive.	1. Select "Book a Ride" . 2. Enter pickup & drop-off locations. 3. Confirm booking.	Pickup: "123 Main St" Dropoff: "456 Park Ave"	Ride booked successfully. Confirmation with driver details. Fare displayed.	Driver R. Silva (Toyota Prius, WP-1234) assigned. Fare Rs. 750 displayed.	Pass
TC - 002	Book Ride Without Login	Test booking when user is not logged in.	User is not logged in.	1. Select "Book a Ride" .	-	User prompted to log in. Redirects to login screen.	System displayed message "Please log in to continue" and redirected to login page.	Pass
TC - 003	Book Ride Without Pickup Location	Test validation when pickup location is missing.	User is logged in.	1. Select "Book a Ride" . 2. Leave pickup field blank. 3. Confirm booking.	Pickup: "" Dropoff: "456 Ave"	Error message: "Enter pickup location" . No booking made.	System displayed error "Pickup location required". Booking not processed.	Pass
TC - 004	Book Ride With Invalid Drop-off	Test validation when drop-off location is invalid.	User is logged in.	1. Select "Book a Ride" . 2. Enter	Pickup: "123 Main" Dropoff: "XYZ123"	Error message: "Invalid drop-off"	System displayed error	Pass

				invalid drop-off. 3. Confirm booking.		location". No booking made.	"Invalid drop-off address". Booking not created.	
TC - 005	Book Ride - Fare Estimate Unavailable	Test behavior when fare estimate cannot be calculated.	User is logged in.	1. Select "Book a Ride" . 2. Enter valid locations. 3. System fails to calculate fare.	Pickup: "123 Main" Dropoff: "456 Ave"	Message displayed: "Fare not available. Try again."	System displayed "Unable to calculate fare. Please try again later."	Pass
TC - 006	Book Ride - Booking Confirmation Failed	Test behavior when booking confirmation fails.	User is logged in.	1. Select "Book a Ride" . 2. Enter valid locations. 3. System fails to confirm booking.	Pickup: "123 Main" Dropoff: "456 Ave"	Error message: "Booking failed. Please try again."	Booking attempt failed. System displayed error "Booking could not be confirmed . Retry."	Pass
TC - 007	Book Ride - App Fails to Load	Test behavior when the booking screen fails to load.	User is logged in.	1. Try to open "Book a Ride" screen.	-	Error message displayed. Prompt user to refresh the page or retry.	App crashed with blank screen, no error message shown	Fail
TC - 008	Book Ride - Cancel After Booking	Test user cancellation after successful booking.	Ride is booked.	1. Go to "My Rides" . 2. Select active ride.	-	Ride canceled successfully. Refund processed (if applicable).	Ride canceled but refund not	Fail

				3. Choose "Cancel Ride" .			processe d to wallet.	
TC - 009	Book Ride - Multiple Payment Methods	Test booking using different payment methods.	User is logged in.	1. Select "Book a Ride" . 2. Choose a payment method. 3. Confirm booking.	Pickup: "A" Dropoff: "B" Payment: Card/Cash	Booking confirmed. Fare paid with selected payment method.	Booking confirmed but only Cash option worked. Card payment failed.	Fail
TC - 0010	Book Ride - Driver Notification	Test notification sent to driver upon ride request.	User is logged in. Driver is available.	1. User books a ride. 2. System assigns a driver.	Pickup: "123 Main" Dropoff: "456 Ave"	Driver receives ride request. Driver details sent to user.	Driver received ride notificatio n instantly. User shown driver name, vehicle, and ETA.	Pass

3.Track a ride.

Test Case ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
TC_01	Track active ride	Verify user can track an ongoing ride.	User has a ride in progress.	1. Open track ride page. 2. Enter ride ID. 3. Click Track.	Ride ID	Map displays ride location in real-time.	(To be filled)	(To be filled)
TC_02	Track completed ride	Verify tracking of a completed ride is possible.	User has a completed ride.	1. Open track ride page. 2. Enter ride ID. 3. Click Track.	Ride ID	Ride details displayed with status "Completed".	(To be filled)	(To be filled)
TC_03	Track non-existent ride	Ensure system handles invalid ride IDs.	None	1. Enter invalid ride ID. 2. Click Track.	Invalid Ride ID	Error message "Ride not found".	(To be filled)	(To be filled)
TC_04	Track ride without login	Verify unauthenticated users cannot track rides.	User is logged out.	1. Open track ride page. 2. Enter ride ID. 3. Click Track.	Ride ID	Redirected to login page.	(To be filled)	(To be filled)

TC_05	Track ride map accuracy	Verify the map shows correct current location.	Ride is in progress.	1. Open track ride page. 2. Track ride.	Ride ID	Map location matches driver's actual location.	(To be filled)	(To be filled)
TC_06	Track ride updates	Verify ride location updates periodically.	Ride is ongoing.	1. Open track ride page. 2. Track ride for 5 min.	Ride ID	Map updates ride location every 10 sec.	(To be filled)	(To be filled)
TC_07	Track ride offline	Check behavior if network is disconnected.	Ride is in progress.	1. Open track ride page. 2. Disable internet. 3. Track ride.	Ride ID	Show error "No internet connection".	(To be filled)	(To be filled)
TC_08	Track ride with invalid format	Verify system handles non-numeric ride IDs.	None	1. Enter non-numeric ID. 2. Click Track.	Ride ID = "ABC123"	Error message "Invalid ride ID format".	(To be filled)	(To be filled)
TC_09	Track ride on mobile	Ensure ride tracking works on mobile.	Ride is in progress.	1. Open track ride page on mobile browser. 2. Enter	Ride ID	Map displays ride correctly on mobile.	(To be filled)	(To be filled)

				ride ID. 3. Click Track.				
TC_10	Track ride history	Verify user can view past ride locations.	User has completed rides.	1. Open ride history page. 2. Click on a ride.	Ride ID	Display ride route and locations.	(To be filled)	(To be filled)

4.Driver Assignment.

Test Case ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
TC_01	Driver assigns for user booked ride.	System assigns nearest available driver	User booked a ride.	1.user booked ride 2.System search	booked a ride from A to B.	Driver assigned successfully.	Driver assigned successfully.	Pass
TC_02	Driver assignment with no available drivers.	System handles no available drivers scenario.	User requests a ride.	1. User searches for a driver 2. System checks available drivers	User requests a ride from Location C to D.	System notifies user of no available drivers.	System notifies users of no available drivers.	Pass
TC_03	Driver assignment for multiple requests.	System handles multiple ride requests from different users.	Multiple users booked rides.	1.Users request rides. 2. System assigns drivers based on proximity	User1: Ride from Location E to F. User2: Ride from Location G to H.	System assigns nearest available drivers to each booking.	System assigns nearest available drivers to each booking.	Pass
TC_04	Driver assignment during peak hours.	System assigns drivers during peak hours.	Multiple users booked rides during peak hours.	1.Users request rides	User requests a ride during	System assigns a driver if available,	System fails to assign a driver due to	Fail

				2. System checks available drivers	peak hours from Location I to J.	otherwise notifies user.	high demand.	
TC_05	Driver assignment with user cancellation.	System handles driver assignment after user cancels ride.	User cancels ride.	1. User cancels ride. 2. System updates driver assignment	Canceled ride from Location K to L.	System notifies driver of ride cancellation.	Driver not notified of cancellation	Fail
TC_06	Driver assignment notification.	System sends notifications to drivers for assigned rides.	A driver is available and has accepted a ride request.	1. User requests a ride. 2. System sends notification	User booked a ride from Location M to N	Driver receives ride assignment notification.	Notification not received by driver	Fail
TC_07	Driver assignment logs.	System logs all driver assignments.	User requests a ride.	1. User requests a ride. 2. System assigns driver. 3. Check logs.	User booked a ride from Location O to P.	Assignment logged correctly in the audit trail.	Assignment not logged in the audit trail.	Fail
TC_08	Driver assignment with user feedback.	System allows users to provide feedback on driver assignment.	User completes ride.	1. User completes ride. 2. User provides feedback.	User booked a ride from Location Q to R.	Feedback is recorded in the system.	Feedback not recorded successfully.	Fail
TC_09	Driver assignment for long distance rides.	System handles long-distance ride assignments.	User requests long distance ride.	1. User requests for long-distance rides. 2. System assigns nearest driver.	Long-distance ride from Location S to T.	Driver is assigned and notified successfully.	Driver not assigned due to system error.	Fail
TC_10	Driver assignment with invalid user input.	System handles invalid input during ride request.	Users enter invalid data.	1. User inputs invalid data. 2. System processes request	Invalid ride request details. Ex: Pickup Location: null	System shows appropriate error message.	System shows unexpected error message.	Fail

					Drop-off Location: B			
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5.Process Payment.

Test Case ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
TC_01	Payment success	Ensures the payment is successfully done with cash or card.	Ride completed	1. Select cash or card 2. If paying using card, enter valid card details 3. Confirm payment	Card details	Payment successful Invoice generated	generated	Pass
TC_02	Invalid card details	Ensure system rejects invalid card numbers, expiry date.	Ride completed, card selected	1. Select card payment 2. Enter invalid card details 3. Confirm payment	Invalid card details	Payment failed, error shown, no invoice generated	no invoice generated	Pass
TC_03	Insufficient balance (Card)	Verify system response when card has insufficient funds.	Ride completed, card selected	1. Select card payment 2. Enter valid card details with low balance 3. Confirm payment	Valid card details with insufficient balance	Payment declined, option to retry or change method	Give pay with cash option	Pass
TC_04	Payment timeout/network error	Ensure the system handles payment gateway timeout.	Ride completed	1. Select card payment 2. Enter valid details 3. Simulate	Valid card details	Payment not processed, retry option shown	Didn't reload	Fail

				network failure or gateway timeout				
TC_05	Duplicate payment attempt	Prevents duplicate charges when user clicks pay twice.	Ride completed	1. Select payment method 2. Enter valid details 3. Click "Pay" button twice quickly	Valid card details	Only one transaction successful, duplicate rejected	Can access one method at time	Pass
TC_06	Cash payment confirmation	Validate cash payment confirmation by driver.	Ride completed	1. Select cash payment 2. Confirm payment received	Cash payment	Payment recorded successfully, invoice generated	Displayed the generated invoice	Pass
TC_07	Partial payment not allowed	System should not accept partial payments.	Ride completed	1. Select card payment 2. Enter amount less than total fare 3. Confirm payment	Partial amount	Error message, full payment required	Only work with card payment	Pass
TC_08	Invoice generation after payment	Ensure invoice is generated with accurate details after payment.	Ride completed, payment successful	1. Complete payment 2. View/download invoice	Valid ride and payment details	Invoice generated with ride ID, fare, payment method, date	Generate automatically with card payments	Pass
TC_09	Invoice not generated on failure	System should not create invoices for failed payments.	Payment attempt fails	1. Attempt payment with invalid/expired card 2. Check for invoice generation	Invalid card details	No invoice generated for failed transaction	Ask to pay with cash	Fail
TC_10	Payment record in history	Verify successful payments are stored in user's	Payment completed successfully	1. Complete ride payment	Ride ID and user	Transaction visible with correct amount,	Share the history with everyone (user,	Pass

		transaction history.		2. Open transaction history	account	date, method, and status	driver and admin)	
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6.Rating and feedback.

Test Case ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
TC_01	Submit rating	User can submit rating for completed ride.	Ride completed.	1. Open ride history. 2. Click Rate. 3. Select stars. 4. Submit.	Ride ID, 5 stars	Rating saved successfully.	Rating submitted successfully.	pass
TC_02	Submit feedback	User can submit feedback for completed ride.	Ride completed.	1. Open ride history. 2. Click Feedback. 3. Enter text. 4. Submit.	Ride ID, Feedback text	Feedback saved successfully.	Feedback saved successfully.	pass
TC_03	Rating without login	Unauthenticated user cannot rate rides.	User logged out.	1. Open ride history. 2. Click Rate.	Ride ID	Redirect to login page.	Redirect to login page.	pass
TC_04	Feedback without login	Ensure feedback submission requires login.	User logged out.	1. Open ride history. 2. Click Feedback.	Ride ID	Redirect to login page.	Redirect to login page.	pass
TC_05	Invalid rating	System rejects invalid ratings.	Ride completed.	1. Select 6 stars. 2. Submit.	Ride ID, Rating = 6	Error "Invalid rating".	Error "Invalid rating"	pass

TC_06	Empty feedback	Ensure feedback cannot be empty.	Ride completed.	1. Click Feedback. 2. Submit empty.	Ride ID, Feedback = ""	Error "Feedback cannot be empty".	(To be filled)	(To be filled)
TC_07	Edit rating	User can edit previous rating.	Rating exists.	1. Open ride history. 2. Edit rating. 3. Submit new rating.	Ride ID, New Rating	Rating updated successfully.	(To be filled)	(To be filled)
TC_08	Edit feedback	User can update feedback.	Feedback exists.	1. Open ride history. 2. Edit feedback. 3. Submit.	Ride ID, New Feedback	Feedback updated successfully.	(To be filled)	(To be filled)
TC_09	Average rating display	Verify driver dashboard shows average rating.	Ratings exist.	1. Open driver dashboard.	Driver ID	Average rating displayed correctly.	(To be filled)	(To be filled)
TC_10	Rating notification	Ensure driver notified when rated.	Ride completed.	1. User submits rating.	Ride ID, Rating	Driver receives notification.	(To be filled)	(To be filled)
Test Case ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
TC_01	Register with valid inputs	Ensure user can log in with correct username and password.	1. User must be registered. 2. User must be in log in page	1. Open the login page. 2. Enter valid name. 3. Enter valid pass. 4. Click Reg.	User name Pass word	User is successfully redirected to the home page	(To be filled after execution)	(To be filled after execution)

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